

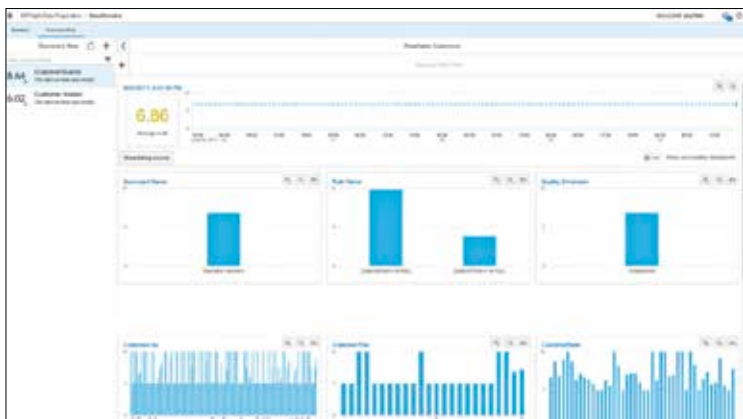
can understand. Remember, beneath every pixel, character, string and number, there is a hidden sequence of binary digits that form the language that the computer uses. Complex problems will have complex datatypes, many of which would differ largely in terms of length, size, frequency or distribution, and can have various interpretations from model to model.

STEPS TO DATA PREPARATION

It all starts from data preparation, where all the relevant attributes in a data string are identified and stored in common file formats such as xml, csv or json. The data is then parsed into a tabular form, for easier scanning and pattern detection. This is then correlated with any external repositories.

Often analysts will find that the data is stored over multiple databases and has to be linked based on matching primary key values. This is where actions such as cross joins and overlaps are used in softwares such as SQL and Tableau to bring multiple files into one input. This is also a crucial point in the data preparation cycle as conditions for conducting sampling tests can be performed as well as any biases in the data can be identified.

Once the data is collected, the next step aims at performing exploratory analysis of any trends, outliers, exceptions, skewed or incorrect records. Having a good eye for what the data is showing is necessary because analysts can assess and remove any unexpected biases. Once the inconsistencies are removed, the rows and columns are formatted based on what the source is. If the data is being extracted from an online server that collects it on a real



Data Visualization